

MOBILITY PRICING INTELLIGENCE: COST PREDICTION & PRICING DRIVER ANALYSIS

Pricing Drivers, Ride Economics &
Mobility Operations



TOOLS USED

Python, Pandas, Numpy, Matplotlib, Plotly &
ML libraries

PRESENTED BY

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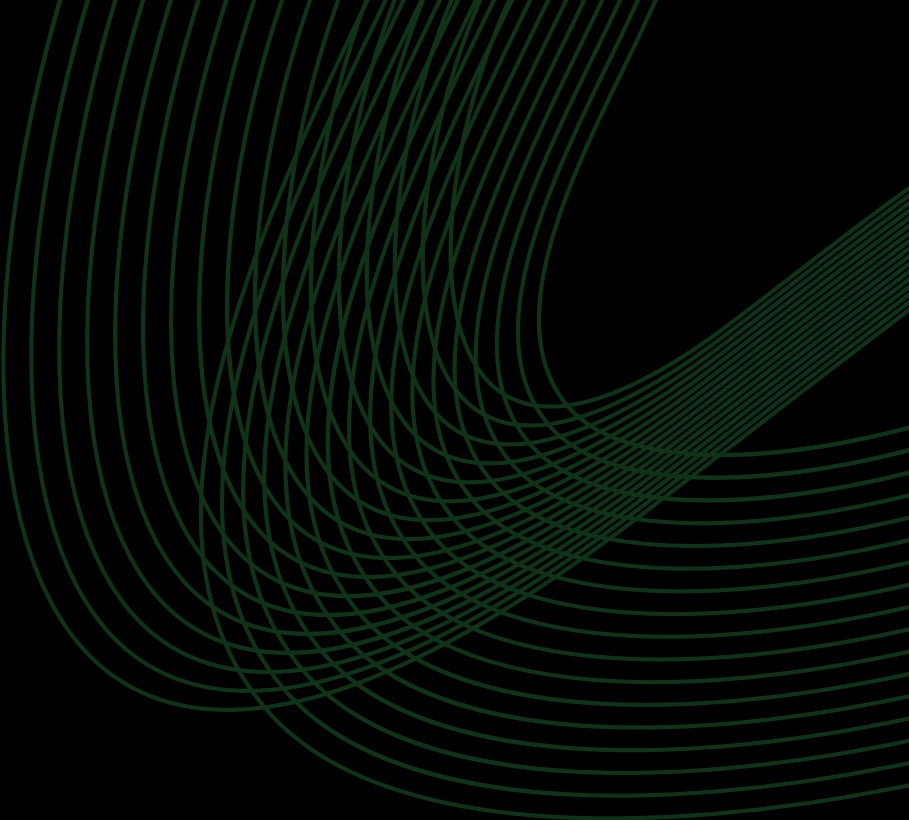
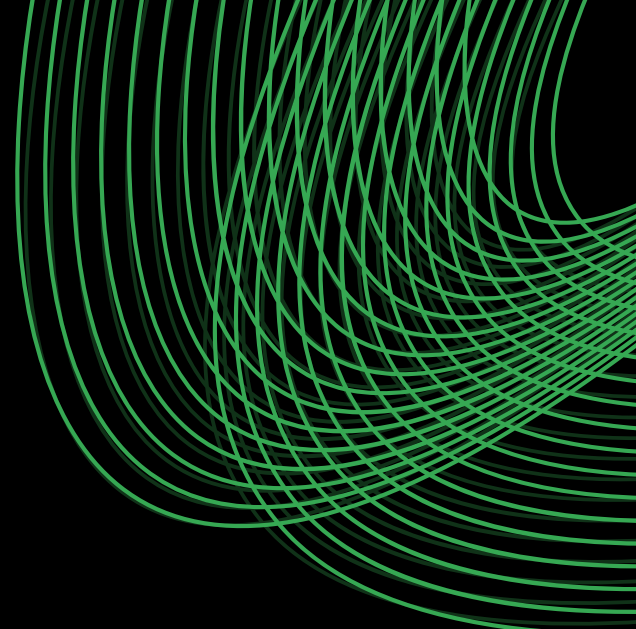


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Overview & Business Objective



1.1 PROJECT OVERVIEW

Ride-sharing platforms operate in an environment where ride costs can vary based on several factors, including expected ride duration, vehicle type, rider demand, driver availability, booking period, location, and customer characteristics.

Understanding these relationships is important for estimating ride costs consistently and identifying the operational factors associated with pricing variation.

The Ride Pricing Analytics project applies machine learning techniques to historical ride-sharing data to develop a ride-cost prediction system and identify the factors that contribute most strongly to model predictions.

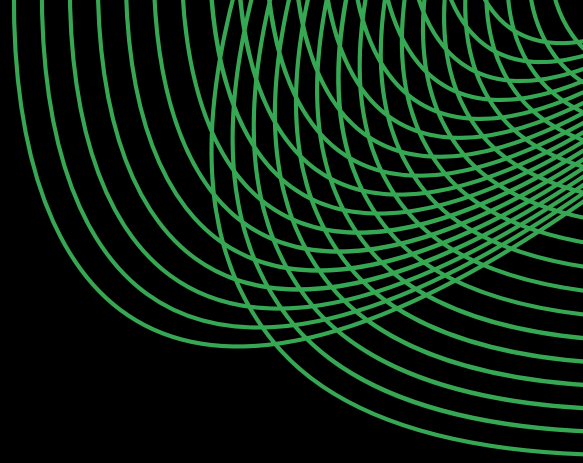
The project combines:

- Data understanding and validation
- Exploratory Data Analysis
- Feature engineering
- Data preprocessing
- Regression modeling
- Hyperparameter tuning
- Ensemble learning
- Model evaluation and comparison
- Prediction-error analysis
- Model explainability
- Business-oriented interpretation

The objective is not only to predict historical ride costs, but also to understand why the models make their predictions and where significant prediction errors remain



1.2 PROBLEM STATEMENT



Ride costs can vary across different combinations of demand, supply, ride duration, vehicle type, booking period, location, and customer characteristics.

The company currently relies primarily on ride duration when determining ride fares. However, the historical dataset contains additional information that may provide useful signals for estimating ride costs.

The key problem addressed in this project is:

How can historical ride-sharing data be used to build a machine learning model that accurately predicts historical ride costs while identifying the major factors contributing to those predictions?

The project therefore treats Historical_Cost_of_Ride as the target variable and uses ride-level, customer, vehicle, booking, and demand-supply information as predictive features.

(Note: It is important to note that the dataset contains historical observed ride costs rather than an explicitly defined optimal or surge-pricing target)



1.3 PROJECT OBJECTIVES

The major objectives of the project are:

- Develop a machine learning-based ride-cost prediction system.
- Understand the underlying structure and characteristics of the data.
- Identify important pricing patterns through EDA.
- Analyze the relationship between ride duration & ride cost.
- Analyze demand and supply relationships using rider and driver information.
- Train and compare multiple regression models.
- Improve selected models using hyperparameter tuning.
- Evaluate Voting and Stacking ensemble approaches.
- Evaluate model performance using appropriate regression metrics.
- Investigate residuals and large prediction errors.
- Identify the most influential features driving model predictions.



1.4 PROJECT SCOPE

The project follows a structured workflow covering data analysis, price modeling, model explainability, and business interpretation to understand demand patterns, evaluate predictive performance, identify key demand drivers, and derive actionable retail insights.

01

Data Analysis

- Dataset inspection
- Data quality assessment
- Distribution analysis
- Relationship analysis

02

Demand Modeling

- Feature preparation
- Regression modeling
- Model comparison
- Model evaluation

03

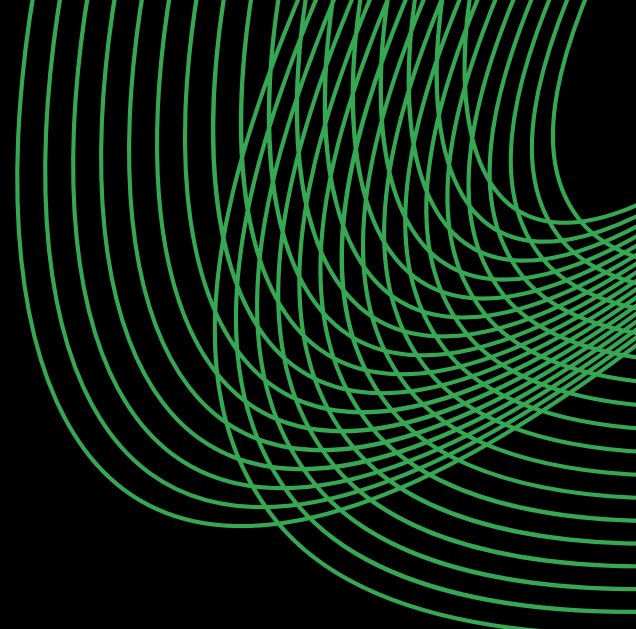
Explainability

- Lasso coefficient analysis
- SHAP analysis
- Permutation feature importance

04

Business Interpretation

- Business Interpretation
- Inventory implications
- Pricing and promotion observations
- Seasonal and weather-related effects
- Recommendations for retail planning



Data Understanding & Exploratory Data Analysis



2.1 DATA OVERVIEW

The dataset contains 1,000 observations and includes information related to riders, drivers, location, customer loyalty, previous rides, ratings, booking time, vehicle type, expected ride duration, and historical ride cost.

Data Source: Kaggle

Dataset Size

1,000

Records

10

Original Features

Purchase Diversity

3

Location Categories

2

Vehicle Types

4

Time Segments



DATA OVERVIEW

Below is a detailed description of the feature set:

Dataset Features	Type	Feature Description
Number_of_Riders	Numerical (Discrete)	Number of riders associated with the observation
Number_of_Drivers	Numerical (Discrete)	Number of available drivers associated with the observation
Location_Category	Categorical	Geographic classification of the ride location
Customer_Loyalty_Status	Categorical	Loyalty category of the customer
Number_of_Past_Rides	Numerical (Discrete)	Number of previous rides completed by the customer
Average_Ratings	Numerical (Discrete)	Average rating associated with the customer
Time_of_Booking	Categorical	Time period during which the ride was booked
Vehicle_Type	Categorical	Category of vehicle selected for the ride
Expected_Ride_Duration	Numerical (Discrete)	Expected duration of the ride
Historical_Cost_of_Ride	Numerical (Continuous)	Historical cost of the ride and target variable used for regression modeling

2.2 TARGET VARIABLE

The target variable selected for prediction is:

Historical_Cost_of_Ride

Historical_Cost_of_Ride represents the observed historical cost associated with each ride.

The target has:

372.50

Mean

362.02

Median

25.99

Minimum

836.12

Maximum

187.16

Standard Deviation

The target demonstrates substantial variation across observations, making it suitable for a continuous regression problem.

